Lecture Scribe Example

097a5e04-f7dd-4595-a859-a8405b48...
PDF

3ada2153-888c-4b6b-9fb3-569b37ed...
PDF

Using the same structure as the example above, generate a lecture scribe for the provided context. The lecture scribe must be written as exam-oriented reference material. Use only the provided lecture slides. Include content only if it appears in the lecture material. Do not add new definitions, examples, explanations, intuition, or interpretations.

Below is the **lecture scribe**, written **strictly using the provided lecture slides** and **following exactly the given scribe structure**.
The content is **exam-oriented**, **reference-style**, and **does not include any additional explanations or interpretations beyond the slides**.

Lecture Scribe

Title

Fundamentals of Probability in Computing

Course Name

CSE400

Lecture Number

Lecture 9

Lecture Topic

Uniform, Exponential, Laplace and Gamma Random Variables
(Focus in slides: Uniform and Exponential Random Variables)

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1. Preliminaries

Definitions Introduced in the Lecture

- **Continuous Random Variable**
- **Uniform Random Variable**
- **Exponential Random Variable**

Notation and Symbols Used

- X: Continuous random variable
- $f_X(x)$: Probability Density Function (PDF)
- $F_X(x)$: Cumulative Distribution Function (CDF)
- a, b: Real constants defining interval
- $u(x)$: Unit step function
- θ: Phase of a sinusoid

Clarification of Symbols Where Required

- $f_X(x) = 0$ outside the specified support
- $F_X(x) = \Pr(X \leqslant x)$

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2. Model Assumptions and Setup

Assumptions Stated in the Lecture

- Random variables are **continuous**
- Probability is defined using **PDF and CDF**
- Support of the random variable is explicitly specified

Conditions Under Which the Framework Applies

- Uniform distribution applies over a **finite interval**
- Exponential distribution applies for $x \geqslant 0$

Setup of Events or Variables

- Uniform RV defined over interval $[a, b]$
- Phase variable θ defined over $[0, 2\pi)$
- Exponential RV defined with parameter $b > 0$

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3. Core Results

Uniform Random Variable

PDF:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leqslant x \leqslant b \\ 0, & \text{elsewhere} \end{cases}$$

CDF:

$$F_X(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leqslant x < b \\ 1, & x \geqslant b \end{cases}$$

Probability Property:

$$\Pr(a < X < b) = \frac{b-a}{\text{length of interval}}$$

Exponential Random Variable

PDF (for $b > 0$):

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

CDF:

$$F_X(x) = [1 - e^{-x/b}]\, u(x)$$

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4. Analytical Development

Uniform RV Probability Calculation

$$\Pr(a < \theta < b) = \int_a^b \frac{1}{2\pi} d\theta$$

Conditional Probability Used

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Trigonometric Condition

$$\cos\theta < \frac{1}{2} \Rightarrow \theta \in (\frac{\pi}{3}, \frac{5\pi}{3})$$

Exponential RV Transformation

$$\Pr(3X < y) = \Pr(X < \frac{y}{3})$$

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5. Illustrative Applications

Example 1: Uniform Random Variable

Given:

$$f_\theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \leqslant \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

(a)

$$\Pr(\theta > 3\pi/4) = \frac{2\pi - 3\pi/4}{2\pi} = \frac{5}{8}$$

(b)

$$\Pr(\theta < \pi \mid \theta > 3\pi/4) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c)

$$\Pr(\cos\theta < 1/2) = \frac{5\pi/3 - \pi/3}{2\pi} = \frac{2}{3}$$

Example 2: Exponential Random Variable

Given:

$$f_X(x) = e^{-x}u(x)$$

(a)

$$\Pr(3X < 5)$$

(b)

$$\Pr(3X < y) \text{ for arbitrary } y$$

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6. Structural Relationships

- Continuous RV → PDF → CDF → Probability
- Uniform RV precedes Exponential RV
- Examples follow immediately after definitions
- Analytical steps use integration and conditional probability
- Applications directly use derived formulas

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7. Consolidated Summary

- Uniform RV has **constant PDF over a finite interval**
- Exponential RV has **decaying PDF for** $x \geqslant 0$
- PDF integrates to obtain probabilities
- CDF used for inequality-based probability queries
- Conditional probability formula applied in examples
- Trigonometric inequalities converted to interval probabilities

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Source: Lecture Slides (CSE400 – Lecture 9)